**MMDU MULLANA OFFICIAL SYLLABUS • BCSE-012 • UNIT 4**

# Unit 4: Cascading Style Sheets (CSS) Architecture (Cascade, Box Model & Layouts)

**Subject:** Fundamentals of Web Technologies (BCSE-012) • **Exam Weightage:** 24 - 28 Marks • **Study Time:** 40 mins

*CSS Cascade & specificity calculation formula matrix; 3 applying methods compared; Typography, line-height & color gradients; Standard CSS box model & universal box-sizing reset; 5 CSS positioning schemes; Floats, micro-clearfix hack & zebra data tables.*

## 1. Cascade, Specificity Weight Matrix & 3 Applying Methods

### 1. Introduction to Style Sheets & Specificity Calculation Formula

Cascading Style Sheets (CSS) describe the visual presentation and formatting of HTML documents, separating structural content from graphic design.

✅ **FUNDAMENTAL LAW / RULE:**
**The CSS Specificity Weight Matrix Formula:**

When conflicting rules target the same element, browsers calculate a 4-part weight:

$$\mathbf{Specificity} = (\text{Inline Style } [1000]) + (\text{IDs } [0100]) + (\text{Classes/Pseudo } [0010]) + (\text{Elements/Tags } [0001])$$

- `!important` overrides standard cascade weights and should be used with extreme caution.

- **Inheritance:** Typography properties ( `color` , `font-family` , `font-size` ) inherit by default to child elements; Box Model properties ( `margin` , `padding` , `border` , `width` ) do NOT inherit.

### 2. Three Methods of Applying CSS

| Method | Syntax Location | Scope | Browser Caching & Reusability |
| --- | --- | --- | --- |
| **Inline CSS** | Inside element tag: | Strictly that single individual element. | **Worst practice.** Pollutes HTML; cannot be cached independently. |
| **Internal / Embedded** | Inside | | |